

STORMY WINDS

The winds generated by global air currents become stronger toward the poles. For example, the westerly winds that blow at latitudes of 40–60° north are much stronger than the tropical trade winds. In the Southern Ocean around Antarctica, this effect generates the powerful winds known as the Roaring Forties, Furious Fifties, and Shrieking Sixties.



▲ FAST AND FURIOUS

The fierce winds of the Furious Fifties make the ocean off the southern tip of South America the roughest in the world.

THE POLAR FRONT

At the polar front (the boundary between a Polar cell and a Ferrel cell), cold polar air pushes beneath the warm air of the Ferrel cell, generating storms that are swept along on the prevailing wind. The northern polar front passes over northern Eurasia and North America, but its position is always changing. When it is north of a particular region, that region enjoys warm, dry weather. When it shifts south, it can bring floods, like these in Slovenia in 2010.



ICY DESERTS

In the polar regions, ice chills the air, making it denser. The dense air sinks and this keeps clouds from forming, so it very rarely snows. In Antarctica and northern Greenland, most of the snow lying on the ground has been there for decades. In the heart of Antarctica, the snow never melts, so there is no liquid water at all, making it the driest desert on Earth.

▼ FREEZE-DRIED

The snow-free Dry Valleys in Antarctica are drier than the Sahara Desert.

